Exact Newton calculus for tangles with a moving meridian

Abstract

We compute the finite two-strand calculus when the main meridian is $M = q^s$, at $q = e^{\pi \mathrm{i}/N}$. Equal and opposite strand orientations have different Casimir relations. Keeping these relations distinct, we prove all six physical quarter-turn formulas, the full pivotal closing functional and the crossing polynomials at colliding roots. The turns reverse fixed Newton coordinates with an explicit paired sine product and an orientation-dependent quadratic scalar. We then contract an original ordered tangle tree without dropping any index or repeated-root derivative, and prove its exact boundary degree and independence of contraction root. One reduced-cut factor survives; the remaining scalar is determined by the physical crossing exponents and writhe. The results hold at every level and on the disk $|s| < 1$. They are finite operator identities, without an asymptotic or volume conclusion.

1 The statement and the objects it evaluates

Fix $N \geq 2$ and write

$$d = N - 1, \quad \theta = \pi/N, \quad q = e^{\mathrm{i}\theta}, \quad Q = q^2, \quad \rho = q - q^{-1}, \quad M = q^s, \quad |s| < 1.$$

All powers q^a mean $\exp(\pi \mathrm{i} a/N)$. A positively directed main strand carries the N-dimensional nilpotent module of highest weight $d + s$; its dual has highest weight $d - s$. We denote them by V_+ and V_-. The parameter is that of the main strand, not the integer color of an auxiliary crossing circle.

A tangle is made from ordered stacking, physical quarter-turns and crossing boxes. The four boundary positions and strand directions are part of its data. A turn bends one incoming and one outgoing strand; it can change an endomorphism into a map between two differently ordered tensor products. After a cylindrical closure giving one knot, we keep one original reduced cut. The zero-framed scalar of this cut operator is denoted $F_N(M)$, with unknot value one.

Define $\mathsf{T}_0 = 2$, $\mathsf{T}_1 = z$ and $\mathsf{T}_{r+1} = z\mathsf{T}_r - \mathsf{T}_{r-1}$. Put

$$P_\sigma(z) = \mathsf{T}_N(z) + 2\cos(2\pi\sigma), \qquad \mathcal{A}_\sigma = \mathbb{C}[z]/(P_\sigma). \tag{1.1}$$

The two equal-color objects use $\sigma = s, -s$; the opposite-color objects use $\sigma = 0$. Though $P_s = P_{-s}$, their ordered physical objects remain distinct. Set

$$S_k = \prod_{r=1}^{k} 2\sin(r\theta), \quad c_j = 2\cos((2j+1)\theta), \quad e_k(z) = S_k^{-2} \prod_{j<k} (c_j - z), \qquad 0 \leq k \leq d. \tag{1.2}$$

Here $S_0 = 1$. These are one fixed basis of polynomials of degree below N, used in each of the different quotients. The functional and the paired products are

$$\phi_\sigma(f) = (-1)^d N^2 [z^d] \operatorname{rem}_{P_\sigma} f, \qquad H_N(s) = \frac{\sin \pi s}{N \sin(\pi s/N)}, \qquad W_k(s) = \prod_{r=1}^{k} \left(1 - \frac{\sin^2(s\theta)}{\sin^2(r\theta)}\right). \tag{1.3}$$

The removable value of H_N at zero is one. H_N and every W_k are nonzero on $|s| < 1$.

Theorem 1.1 (Exact moving-meridian calculus)**.** *The actual pair Casimir satisfies its relation* (1.1)*, and the physical partial pivotal closure of a polynomial is $H_N(s)\phi_\sigma(f)$ times the identity on the surviving strand. In fixed coordinates,*

$$\phi_\sigma(e_i e_j) = \mathbf{1}_{i+j=d}.$$

Every physical forward turn is

$$\mathscr{T}_s^{\mathrm{fix}} = \frac{\gamma(s)}{H_N(s)} \mathcal{R}\,\mathrm{diag}(W_0(s), \dots, W_d(s)), \qquad \mathcal{R}e_k = e_{d-k}, \tag{1.4}$$

where the six domain/codomain types and their scalars $\gamma \in \{1, q^{-s^2/2}, q^{s^2/2}\}$ are given in Table 1. The crossing, its inverse and every fixed integer power have holomorphic canonical polynomial representatives on this disk. At repeated roots these representatives retain their first polynomial derivatives.

For an original ordered tree contraction, the corresponding edge kernel is

$$K_N(t, u; s) = \sum_{k=0}^{d} e_k(-2\cos t) e_k(-2\cos u) \frac{H_N(s)}{W_k(s)}. \tag{1.5}$$

The exact recursion of Theorem 7.1 evaluates the whole original contraction. Its output to a parent has trigonometric degree at most d, independently of the size of the subtree. Any root can be used for this finite summation, provided all original tensors and reference maps are retained.

The theorem supplies the finite adapter needed when the meridian moves. It does not bound the signed contraction by a positive complete-color sum. Nor does it replace the two quotient multiplications by one: products which vanish in $\mathcal{A}_0$ can return in $\mathcal{A}_s$.

We use the complete calculations of [1] only where the color is exactly complete: Proposition 3.2 for the physical turn and closing calibration, and equation (4.8) for the complete crossing polynomial. The physical normalization at $M = 1$, for $N \geq 3$, is [2, Theorem 10.1]. Canonical monic reduction and its compatibility with products are [2, Propositions 2.1–2.2]; we also give the needed coefficient arguments below. The moving formulas, their orientation scalars and their contraction are proved here. Nilpotent colored tangles and their ribbon interpretation go back to Akutsu–Deguchi–Ohtsuki [3]; the unrolled description and duality conventions are discussed in [4]. These references provide the construction's background, not a substitute for the moving-coordinate calculations.

2 Physical modules, bending maps and the cut

For a highest root $x = q^\alpha$, let V_x have basis $v_0, \dots, v_d$ and matrices

$$\begin{aligned} Kv_a &= xQ^{-a}v_a, & Fv_a &= v_{a+1}, & Ev_a &= \varepsilon_a(x)v_{a-1}, \\ \varepsilon_a(x) &= [a]\frac{xq^{1-a} - x^{-1}q^{a-1}}{\rho}, & [a] &= \frac{q^a - q^{-a}}{\rho}. \end{aligned} \tag{2.1}$$

Missing endpoint vectors are zero. We use

$$\Delta E = E \otimes K + 1 \otimes E, \quad \Delta F = F \otimes 1 + K^{-1} \otimes F, \quad \Delta K = K \otimes K.$$

The antipode is $S(E) = -EK^{-1}$, $S(F) = -KF$, $S(K) = K^{-1}$. For $x = -q^{-1}q^s = q^{d+s}$, its dual root is $y = x^{-1}Q^{-1} = q^{d-s}$, and $\varepsilon_a(x) = [a][a-s]$, $\varepsilon_a(y) = [a][a+s]$. They are nonzero for $1 \leq a < N$ and $|s| < 1$, at either parity.

Write φ_a for the dual basis of V_x. The identification with its actual dual is

$$J_x : V_y \longrightarrow V_x^*, \qquad J_x w_a = \gamma_a(x)\varphi_{d-a}, \qquad \gamma_a(x) = (-x)^a Q^{a(a+1)/2}. \tag{2.2}$$

For example $\gamma_{a+1}/\gamma_a = -xQ^{a+1}$, and substitution in the antipode actions checks E, F, K separately. The ordinary right evaluation is $\varphi(v)$ and the coevaluation is $\sum_a v_a \otimes \varphi_a$. The left evaluation is $\varphi(g_x v)$ and the left coevaluation is $\sum_a \varphi_a \otimes g_x^{-1} v_a$, where

$$(J_x^{-1})^{\mathsf{t}} J_y = g_x = K^{1-N}. \tag{2.3}$$

Indeed its ath entry is $\gamma_a(y)/\gamma_{d-a}(x) = x^{1-N}Q^{a(N-1)}$. Inserting these sums proves all four snake identities. Thus a bend and its inverse leave neither a missing pivot nor a scalar.

For weights α, β, put $Hv_a = (\alpha - 2a)v_a$ on the first factor and similarly on the second. The full crossing is

$$\check{R}_{x,y} = P\mathscr{R}_{x,y}, \qquad \mathscr{R}_{x,y} = q^{H\otimes H/2} \sum_{k=0}^{d} \frac{q^{k(k-1)/2}\rho^k}{[k]!} E^k \otimes F^k, \qquad [k]! = \prod_{r=1}^{k} [r]. \tag{2.4}$$

Here P exchanges the two tensor factors; the coefficient is never stripped of its Cartan factor. The finite denominators are nonzero. The usual intertwining identities can also be checked directly: commuting one raising or lowering matrix through the kth term pairs it with the $(k-1)$st, whose coefficient ratio is $q^{k-1}\rho/[k]$; the two end terms vanish by $E^N = F^N = 0$. This is the finite nilpotent ribbon crossing. We retain its actual matrix inverse for negative crossings.

A partial closure with g_y of a pair intertwiner is again an intertwiner on the open factor. Here is an elementary verification useful later. For $Z = E$ or KF, $\Delta Z = Z \otimes K + 1 \otimes Z$. Multiply the intertwining identity by $1 \otimes K^{-1} g_y$ before tracing the second factor. The first terms give the commutator with Z on the first factor; the remaining terms cancel by cyclicity because $K^{-1} g_y = K^{-N}$ is central. The K equation gives commutation with K and hence with F. The single module is simple on our disk: its distinct K weights and nonzero arrows connect all basis vectors. Thus the closure is a scalar.

In particular the positive curl is obtained by closing $\check{R}_{x,x}$. Evaluate on v_0: only the zero-transfer term with closed index zero returns to that coefficient. Its value is $q^{\alpha^2/2}x^{1-N}$. For the negative curl evaluate the inverse on v_d; again just the extreme zero-transfer term remains, with value $q^{-(\alpha-2d)^2/2}(xQ^{-d})^{1-N}$. Consequently

$$\vartheta_\alpha = q^{(\alpha^2 - 2d\alpha)/2}, \qquad \vartheta_{d\pm s} = q^{(s^2-d^2)/2}, \qquad \frac{\vartheta_{d+s}^{-w}}{\vartheta_d^{-w}} = q^{-ws^2/2}. \tag{2.5}$$

This derives the framing factor in the same physical convention.

The reduced cut may be implemented by any rank-one projector $e = \psi\varphi$ with $\varphi(\psi) = 1$: with defect $\mathcal{D} = g^{-1}e$, a scalar cut operator $C = cI$ gives $\mathrm{Tr}(g\mathcal{D}C) = \mathrm{Tr}(eC) = c$. We keep this one cut, rather than closing the last strand by its zero ordinary quantum dimension. Theorem 10.1 of [2], together with the complete basis calibration below, identifies the complete value with the reduced Kashaev invariant for $N \geq 3$. The finite operator calculus here also holds at $N = 2$, without invoking that comparison there. Nothing in our calculation divides by that value; it may be zero at a particular finite level.

3 The Casimir relation and the full pivotal moments

On $V_x \otimes V_y$ define

$$\Omega = \rho^2 F_{12}E_{12} + qK_{12} + q^{-1}K_{12}^{-1} = \rho^2 E_{12}F_{12} + qK_{12}^{-1} + q^{-1}K_{12}. \tag{3.1}$$

It commutes with the three total generators. For arbitrary nonzero roots x, y, let

$$P_{x,y}(z) = \prod_{j=0}^{d} (z - qxyQ^{-j} - q^{-1}(xy)^{-1}Q^j).$$

Proposition 3.1 (Polynomial and moment identities)**.** *For all nonzero x, y, as identities of the finite matrices,*

$$P_{x,y} = \mathsf{T}_N + (xy)^N + (xy)^{-N}, \qquad P_{x,y}(\Omega) = 0. \tag{3.2}$$

The actual partial pivotal moments are

$$\operatorname{ptr}_{g_y}(\Omega^m) = 0 \quad (0 \le m < d), \qquad \operatorname{ptr}_{g_y}(\Omega^d) = s_x I, \qquad s_x = -N\frac{x^N - x^{-N}}{qx - q^{-1}x^{-1}}. \tag{3.3}$$

The apparent singularities in s_x are removable Laurent polynomial values. For $x = q^{d\pm s}$, $s_x = s_0 H_N(s)$ with $s_0 = (-1)^d N^2$.

Proof. First choose generic colors so all roots and relevant raising factors are distinct and nonzero. In total degree j, a highest vector $h_j = \sum_{a=0}^{j} b_a v_a \otimes w_{j-a}$, $b_0 = 1$, is determined by

$$\frac{b_{a+1}}{b_a} = -\frac{\varepsilon_{j-a}(y)}{\varepsilon_{a+1}(x) y Q^{-(j-a-1)}}.$$

Its highest root is xyQ^{-j}. The commutation relation shows that its first N lowering vectors are nonzero: a shorter string would contradict a nonzero raising factor. The root binomial identity for the two Q-commuting summands of F_{12} gives $F_{12}^N = 0$; the same holds for E_{12}. The N distinct Casimirs therefore give N independent summands of dimension N, exhausting the pair. This proves annihilation generically. Every matrix entry is Laurent in x, y, so it proves the identity at all their specializations. Putting $z = w + w^{-1}$ in the monic root product proves its Chebyshev expression, since $(qxy)^N = -(xy)^N$.

For the moments, the preceding partial-closure argument makes the answer scalar at generic colors. Compute its first diagonal entry. In the total-degree block through $v_0 \otimes w_\ell$, put $t = Q^{-\ell}$ and use first index a. The three transition entries of Ω are

$$\begin{aligned} D_a(t) &= x^{-1}Q^a(qy + q^{-1}y^{-1}) \\ &\quad + yQ^a(qx + q^{-1}x^{-1} - (q + q^{-1})x^{-1}Q^a)t, \\ a &\longrightarrow a+1 : \rho^2 \varepsilon_{\ell-a}(y), \\ a &\longrightarrow a-1 : \rho^2 \varepsilon_a(x) x^{-1} y Q^{2a-1} t. \end{aligned} \tag{3.4}$$

The increasing entry has powers $-1, 0, 1$ in t, and the decreasing entry has power one. A returning path has equally many of each; hence $H_m(t) = (\Omega^m)_{0,\ell;0,\ell}$ has degrees from zero to m. At a forbidden lower endpoint the factor ε_0 kills the path; fewer than N steps cannot encounter a spurious upper endpoint. Thus this one polynomial describes every ℓ.

The pivot contributes $y^{1-N}Q^{-\ell}$. Summing $\ell = 0, \ldots, d$ kills every power of $tH_m(t)$ except multiples of N. This gives the lower moments and, for $m = d$, selects only the leading coefficient. Removing its factor y^m gives the tridiagonal matrix

$$L_{aa} = Q^a(qx + q^{-1}x^{-1} - (q + q^{-1})x^{-1}Q^a), \quad L_{a+1,a} = -qQ^a, \quad L_{a-1,a} = \rho^2 \varepsilon_a(x) x^{-1} Q^{2a-1}.$$

Its first column has the exact powers

$$(L^m)_{a0} = (-1)^a q^m Q^{a(a-1)/2} \begin{bmatrix} m \\ a \end{bmatrix}_Q \prod_{r=a}^{m-1} (x - Q^r x^{-1}), \quad 0 \le a \le m < N. \tag{3.5}$$

Here $\begin{bmatrix} m \\ a \end{bmatrix}_Q = (Q;Q)_m/((Q;Q)_a(Q;Q)_{m-a})$ and $(v;Q)_k = \prod_{r<k}(1-vQ^r)$. To verify the formula, multiply one column by the three displayed entries. For the x coefficients use $\begin{bmatrix} m+1 \\ a \end{bmatrix}_Q = \begin{bmatrix} m \\ a-1 \end{bmatrix}_Q + Q^a \begin{bmatrix} m \\ a \end{bmatrix}_Q$; for x^{-1} use also $(1-Q^{a+1})\begin{bmatrix} m \\ a+1 \end{bmatrix}_Q = (1-Q^{m-a})\begin{bmatrix} m \\ a \end{bmatrix}_Q$. After these substitutions the two coefficients equal the product in (3.5) at $m+1$; the end indices have their single surviving terms. The identity at $m=0$ starts the induction. For $m=d, a=0$ the product is

$$q^d \prod_{r=0}^{N-2} (x - Q^r x^{-1}) = -\frac{x^N - x^{-N}}{qx - q^{-1}x^{-1}}.$$

Root orthogonality supplies the factor N, and $y^{1-N}y^d = 1$. This proves (3.3) and then its specializations by Laurent continuation. Substituting $x = q^{d\pm s}$ gives the stated $s_0 H_N$. □

Lemma 3.2 (Pointwise pair algebra)**.** *On $|s| < 1$, for each of the four ordered sign objects, the complete commutant of the pair generators is its rank-N Casimir algebra (1.1). This includes $s = 0, \pm 1/2$.*

Proof. Let W_T be the span of $v_a \otimes w_{T-a}$. Other K_{12} eigenspaces can combine W_T and W_{T+N}, but W_d alone has its eigenvalue: no other total in $0, \dots, 2d$ differs from d by a nonzero multiple of N. For $T < d$, the first $T+1$ columns of $E_{12} : W_{T+1} \to W_T$ give a triangular maximal minor with nonzero second-leg factors of indices $1, \dots, T+1$. For $T \geq d$, a triangular maximal minor of $F_{12} : W_T \to W_{T+1}$ has unit first-leg factors. These maps are onto. Thus W_d generates all blocks by raising and lowering.

The middle Casimir is tridiagonal and each adjacent entry is a nonzero unit times $[r][r \pm s]$, $1 \leq r < N$. Its boundary vector is cyclic. Its commutant is therefore the polynomials of degree below N: the image of that cyclic vector determines a commuting operator. A pair intertwiner preserves W_d and commutes with this block; subtract its polynomial in Ω. The difference is zero on W_d, hence on the generated pair. The annihilator in Proposition 3.1 has degree N, so it is precisely the minimal polynomial. No specialization-of-dimension argument is used. □

It follows for every polynomial f that

$$\mathrm{ptr}_{g_y} f(\Omega) = s_x [z^d] \operatorname{rem}_{P_{x,y}} f \; I. \tag{3.6}$$

Left closure gives the same formula with the surviving color's s_y. Indeed dualize the displayed calculation: $S(\Omega) = \Omega$, the left pivot becomes g_x^{-1}, and $s_{x^{-1}Q^{-1}} = s_x$. On the dual pair $s_x = s_y = s_0 H_N$. This proves the actual closing part of Theorem 1.1.

The right and left partial closures of the same-color pair agree, since the surviving factor has the same s_x in both. The same is true after dualizing a strand, because $s_y = s_x$. To compare two possible reduced cuts of a one-component tangle, keep both cuts open and use the evaluations (2.2) to put its four ends in a pair endomorphism. Lemma 3.2 writes this operator as $f(\Omega)$. Closing the first cut or the second gives the respective partial closures just computed; they are the same scalar operator. This proves independence of the reduced cut, including at repeated pair roots. It uses no division by an ordinary quantum dimension. The ribbon tangle construction in [3, 5] supplies the usual Reidemeister and composition framework; the crossing, bends, pivot, framing and cut normalization used in that construction have all been specified here.

4 Newton coordinates and repeated roots

It is useful to distinguish a basis adapted to the moving nodes from our fixed basis. Write

$$c_j(s) = 2\cos((2j+1-2s)\theta), \qquad e_k^s(z) = S_k^{-2} \prod_{j<k} (c_j(s) - z).$$

The reflection is $c_{d-j}(s) = c_j(-s)$. The elementary product $S_i S_{d-i} = N$ follows from $\prod_{r=1}^{N-1} 2\sin(r\theta) = N$.

Lemma 4.1 (Two exact dualities)**.** *In the indicated quotient algebras,*

$$\phi_s(e_i^s e_j^{-s}) = \mathbf{1}_{i+j=d}, \qquad \phi_\sigma(e_i e_j) = \mathbf{1}_{i+j=d}. \tag{4.1}$$

Thus the fixed coefficient of e_i in a canonical polynomial f is $\phi_\sigma(f e_{d-i})$.

Proof. For the moving bases, a product of degree below d has zero top coefficient. At degree d its leading coefficient is $(-1)^d/N^2$. If $i+j \geq N$, the first i nodes and the reflected last j nodes cover all N nodes, with their multiplicities, so the product is divisible by P_s. This proves the first identity.

For the second, put $\Delta = 2\cos(2\pi\sigma) - 2$. Divide the fixed product by P_0 as $e_i e_j = P_0 A + B$, $\deg B < N$. Since its degree is at most $2N-2$, $\deg A \leq N-2$. Its remainder modulo $P_\sigma = P_0 + \Delta$ is $B - \Delta A$, whose top coefficient equals that of B. Apply the first identity at zero. This argument explains why the pairing stays fixed although multiplication changes. □

For example, if $i + j \geq N$ then $e_i e_j = P_0 A$ and

$$\mathrm{rem}_{P_s}(e_i e_j) = -\big(2\cos(2\pi s) - 2\big)A.$$

The right side need not vanish. A coordinate reversal is a linear map between the relevant polynomial spaces, not an identification of their products.

The trace also gives a division-free way to recover the polynomial of any physical intertwiner $A(s)$ on one ordered pair:

$$[e_i]A(s) = \frac{1}{H_N(s)}\,\mathrm{ptr}_g\big(A(s)e_{d-i}(\Omega)\big). \tag{4.2}$$

The scalar on the right is read on the surviving strand. Lemma 3.2 and (4.1) prove the identity. In particular the coefficients of a specified holomorphic physical family are holomorphic throughout $|s| < 1$. Colliding pair eigenvalues do not make this extractor singular.

For clarity we record what the functional means at a collision. For a canonical F and a monic P of degree N, $[z^d]F$ is the sum of the residues of $F(z)/P(z)$ at its finite poles. If a is simple its contribution is $F(a)/P'(a)$. If $P = (z-a)^2 Q_a$ its contribution is

$$\frac{F'(a)}{Q_a(a)} - \frac{F(a)Q_a'(a)}{Q_a(a)^2}. \tag{4.3}$$

Multiply their sum by $(-1)^d N^2$ to obtain ϕ. The residue statement follows by expanding F/P at infinity: its z^{-1} coefficient is the top coefficient of the remainder. It also proves that replacing a polynomial by its remainder preserves these complete data. At $s = 0$ every root is double except the simple root -2 when N is odd. At $s = \pm 1/2$ the endpoint roots which occur are simple and the other roots are double. Thus (4.3) specifies every normal that is needed below.

5 The physical quarter-turn

We now compute the turn itself. It is helpful to keep three kinds of indices separate: e_k is a polynomial, $\mathbf{e}_j$ below is a coordinate vector in a pair module, and a, b in a matrix entry are physical strand colors.

For $A : V_{x_1} \otimes V_{x_2} \to V_{y_1} \otimes V_{y_2}$, bending the first output and second input gives a map $V_{y_1}^* \otimes V_{x_1} \to V_{y_2} \otimes V_{x_2}^*$. Inserting (2.2) in evaluation and coevaluation gives

$$(\mathscr{T}A)^{jb}_{ak} = \frac{\gamma_a(y_1)}{\gamma_b(x_2)} A^{d-a,j}_{k,d-b}. \tag{5.1}$$

In this formula the arguments of γ are highest roots; the subscripts are basis indices. In particular the denominator is $\gamma_b(x)$ for an xx endomorphism, and $\gamma_b(y)$ for an xy endomorphism. The formula fixes which forward turn we mean.

Put $A = V_x \otimes V_y$, $B = V_y \otimes V_x$. For morphisms between them use the carriers

$$G_A = I, \qquad G_B = \check{R}_{x,y} r_0(\Omega_A)^{-1}, \qquad C_{T,S} = G_T G_S^{-1}. \tag{5.2}$$

Here r_0 is the complete crossing polynomial, including its full normal data; its nodal values are nonzero, so it is a unit. For the equal-color objects the carrier is the identity. A morphism $S \to T$ is written $C_{T,S} f(\Omega_S)$, in this order. The Casimir intertwines with each carrier. Therefore composition is the product of the polynomials in the source quotient, with no scalar left between adjacent carriers. At $s = 0$, $G_B = I$ after the complete physical identification.

5.1 A three-strand calculation with all channels retained

Use the fixed balanced gauge

$$|a\rangle = \delta_a v_a, \quad \delta_a = \rho^a/(Q;Q)_a.$$

The finite identity $(Q;Q)_a(Q;Q)_{d-a} = N(-1)^a Q^{a(a+1)/2}$ gives

$$\delta_a \delta_{d-a} \gamma_a(-q^{-1}) = \frac{\rho^d}{N}(-q^{-1})^a.$$

Consequently (5.1) agrees with the complete balanced turn of [1, Proposition 3.2]. This fixes the physical calibration rather than allowing an arbitrary turn scalar.

A vector $w \in W_d \subset V_x \otimes V_x$ determines an intertwiner $I_w : V_x \to V_x \otimes V_x \otimes V_y$ by the highest vector

$$H_L(w) = \sum_{c=0}^{d} \frac{(-y)^c Q^{-c(c-1)/2}}{\prod_{r=1}^{c} \varepsilon_r(y)} E_{12}^c w \otimes w_c. \tag{5.3}$$

Indeed its highest-vector equation says $z_{c+1} = -yQ^{-c}E_{12}z_c/\varepsilon_{c+1}(y)$; its last condition is $E_{12}^N w = 0$. Its highest root is $x^2yQ^{-d} = x$, and its F-string defines the intertwiner on all of V_x. Conversely, third color zero in a highest vector of this weight forces first-pair degree d: the other congruent degree $2N-1$ is outside the pair. The recursion then determines every coefficient. Hence (5.3) describes the whole N-dimensional Hom space, including repeated-channel directions.

Let $\mathbf{e}_j = \delta_j \delta_{d-j} v_j \otimes v_{d-j}$ in W_d. The two pair Casimirs act on its coordinates as A_s and C_s, respectively. Set

$$P_j(s) = \prod_{r=1}^{j} \frac{\sin((r-s)\theta)}{\sin(r\theta)}, \qquad \mathsf{D}_s = \operatorname{diag}(P_0(s), \dots, P_d(s)).$$

Extracting third color zero from (5.3) gives

$$\begin{aligned}
\mathsf{C}_s &= \mathsf{D}_s^{-1} \mathsf{C}_0 \mathsf{D}_s, \\
(\mathsf{C}_0)_{jj} &= c_j, & (\mathsf{C}_0)_{j-1,j} &= \rho^2[j]^2, \\
(\mathsf{A}_s)_{jj} &= 4qQ^j - q(1+Q)Q^{2j} + \rho^2[s]^2 qQ^j, & & \\
(\mathsf{A}_s)_{j-1,j} &= -\rho^2 Q^j [j][j-s], & (\mathsf{A}_s)_{j,j-1} &= -\rho^2 Q^j [j][j+s].
\end{aligned} \tag{5.4}$$

Only third colors zero and one contribute to the first extraction. Their F_2E_2 terms cancel, leaving $-\rho^2 y F_2 E_1 K_2$ and the two Cartan terms. These give the diagonal c_j and superdiagonal $\rho^2[j][j-s]$ displayed above. The first-pair entries follow directly from (3.1). Using $[j-s][j+s] = [j]^2 - [s]^2$ then gives

$$\mathsf{D}_s \mathsf{A}_s \mathsf{D}_s^{-1} = \mathsf{A}_0 + \rho^2[s]^2 \mathsf{K}, \qquad \mathsf{K}_{jj} = qQ^j, \quad \mathsf{K}_{j,j-1} = Q^j. \tag{5.5}$$

Let $U = r_0(\mathsf{C}_0)$. The complete polynomial $r_0 = \kappa_0 \sum_{n=0}^{d} (-q)^n e_n$, $\kappa_0 = q^{d^2/2}$, gives by bidiagonal multiplication

$$U_{ij} = \kappa_0 (-1)^j q^{j+i^2} \begin{bmatrix} j \\ i \end{bmatrix}_Q \quad (i \le j), \qquad U_{ij} = 0 \quad (i > j). \tag{5.6}$$

There are two conjugations which we will need:

$$U^{-1} \mathsf{A}_0 U = \mathsf{C}_0^{\mathsf{t}}, \qquad U^{-1} \mathsf{K} U = -\mathsf{S}_{\mathrm{low}} + (-1)^N \mathbf{e}_0 \mathbf{e}_d^{\mathsf{t}}. \tag{5.7}$$

Here $\mathsf{S}_{\mathrm{low}} \mathbf{e}_j = \mathbf{e}_{j+1}$ for $j < d$ and its last column is zero. To check the first identity, the three entry ratios in $\mathsf{A}_0 U = U \mathsf{C}_0^{\mathsf{t}}$ are

$$\frac{U_{i-1,j}}{U_{ij}} = q^{-2i+1} \frac{1-Q^i}{1-Q^{j-i+1}}, \quad \frac{U_{i+1,j}}{U_{ij}} = q^{2i+1} \frac{1-Q^{j-i}}{1-Q^{i+1}}, \quad \frac{U_{i,j+1}}{U_{ij}} = -q \frac{1-Q^{j+1}}{1-Q^{j-i+1}}.$$

Substitute (5.4) and clear the denominators; the terms agree after $Q = q^2$. There is one boundary with a vanishing cleared denominator, namely $(i,j) = (0,d)$; it needs a direct check. Here $U_{1d}/U_{0d} = -q^{-1}$, and

$$(\mathsf{A}_0)_{00} + (\mathsf{A}_0)_{01} U_{1d}/U_{0d} = q(3-Q) + (Q-1)^2 q^{-1} = q + q^{-1} = c_d.$$

For $i = j+1$, the required equality is $U_{ii}/U_{jj} = -Q^i$; for $i > j+1$ both sides vanish. The other endpoint terms have their ordinary zero factors $[0] = [N] = 0$. For the second identity, column $j < d$ reduces to $Q^i \begin{bmatrix} j \\ i \end{bmatrix}_Q + \begin{bmatrix} j \\ i-1 \end{bmatrix}_Q = \begin{bmatrix} j+1 \\ i \end{bmatrix}_Q$ and is the negative of column $j+1$ of U. In the last column $\begin{bmatrix} N \\ i \end{bmatrix}_Q = 0$ for $1 \le i < N$; its zeroth entry is $(-1)^N \kappa_0$. This proves the additional last-column term, which must not be discarded at a moving meridian.

Let $J\mathbf{e}_j = \mathbf{e}_{d-j}$ and $\mathsf{V}_s = JU^{-1}\mathsf{D}_s$. Equations (5.4)–(5.7) imply

$$\mathsf{V}_s \mathsf{C}_s \mathsf{V}_s^{-1} = \mathsf{C}_0^{\mathsf{t}}, \qquad \mathsf{V}_s \mathsf{A}_s \mathsf{V}_s^{-1} = \mathsf{L}_s, \tag{5.8}$$

where the only entries of L_s are

$$(\mathsf{L}_s)_{kk} = c_k, \quad (\mathsf{L}_s)_{k-1,k} = -4\sin((k-s)\theta)\sin((k+s)\theta), \quad (\mathsf{L}_s)_{d,0} = (-1)^{N+1} 4\sin^2(s\theta).$$

For $N = 2$ the superdiagonal and bottom-left entries occupy different positions, as they should. The wrap entry changes the characteristic polynomial from P_0 to P_s.

5.2 Identifying these coordinates with a physical bend

The endpoint vector in our gauge is $\mathbf{e}_d = \delta_d v_d \otimes v_0$. Therefore

$$H_L(\mathbf{e}_d) = \delta_d \sum_{c=0}^{d} (-1)^c Q^{-c(c+1)/2} v_{d-c} \otimes v_0 \otimes w_c. \tag{5.9}$$

Only the zero-transfer term of the crossing on the last pair acts on v_0. Its coefficient is $\kappa_{xy} q^{-(d+s)c}$, $\kappa_{xy} = q^{(d^2-s^2)/2}$. Together with (5.9) this is $\delta_d \kappa_{xy} \gamma_c(x)^{-1}$. Thus, as intertwiners,

$$\check{R}_{23} I_{\mathbf{e}_d} = \delta_d \kappa_{xy} (\mathrm{coev}_{x,y} \otimes I_x).$$

Insert f in the first pair and apply the snake identity. The gauged vector $\mathbf{e}_0 = \delta_d v_0 \otimes v_d$ has the same endpoint factor, so the resulting identity is exactly

$$[\mathscr{T}_x(f)\check{R}_{x,y}]_{23} I_{\mathbf{e}_d} = \kappa_{xy} I_{f(\mathsf{A}_s)\mathbf{e}_0}. \tag{5.10}$$

The common δ_d has canceled; it has not been assigned to the crossing or omitted from one endpoint.

The finite sine product gives

$$P_d(\pm s) = H_N(s), \qquad P_k(-s)P_{d-k}(s) = H_N(s), \qquad W_k W_{d-k} = H_N^2. \tag{5.11}$$

For example the first follows by factoring $\sin \pi s = N \sin(s\theta) \prod_{r=1}^{d} \sin((r-s)\theta)/\sin(r\theta)$; reflecting the complementary indices proves the other two. The bidiagonal matrix satisfies $e_k(\mathsf{C}_s)\mathbf{e}_d = P_k(-s)\mathbf{e}_{d-k}$. Write $\widehat{\mathscr{T}_s}(f) = \mathscr{T}_x(f)G_B$ in fixed coordinates. Since $r_0(\mathsf{C}_s) = \mathsf{D}_s^{-1} U \mathsf{D}_s$, (5.10) and (5.11) give

$$\operatorname{coeff} \widehat{\mathscr{T}_s}(f) = \frac{\kappa_{xy}}{H_N} \mathsf{V}_s f(\mathsf{A}_s)\mathbf{e}_0 = \frac{q^{-s^2/2}}{H_N} f(\mathsf{L}_s)\mathbf{e}_d. \tag{5.12}$$

Here $\mathsf{V}_s \mathbf{e}_0 = \kappa_0^{-1}\mathbf{e}_d$. On e_n, the diagonal cancels successively because $c_{d-n} = c_n$. The unique remaining path moves upward n times and gives

$$e_n(\mathsf{L}_s)\mathbf{e}_d = W_n(s)\mathbf{e}_{d-n}.$$

The wrap entry cannot act in fewer than N steps. This proves the first turn in Table 1 for the whole intertwiner, not just for an extreme matrix coefficient.

5.3 Opposite colors and all six scalars

On total degree $t \le d$ of an opposite pair the Casimir has characteristic and minimal polynomial $\prod_{j=0}^{t}(z - c_j)$. Generic highest vectors give the characteristic identity; Laurent continuation preserves it. Nonzero adjacent entries make the block cyclic even at repeated roots. Consequently e_n is zero on that block if $n > t$, has zero extreme entry if $n < t$, and for $n = t$ its unique extreme path gives

$$(e_t(\Omega_{x,y}))_{(0,t);(t,0)} = M^{-2t} P_t(s). \tag{5.13}$$

Before the leading coefficient $(-1)^t/S_t^2$, the path is $\rho^{2t}(y/x)^t \prod_{a=1}^{t} \varepsilon_a(x)$; its Q powers cancel, proving the formula.

Apply (5.1) to this endomorphism. Bandwidth makes its turn vanish on target blocks below degree $d-n$, so its polynomial is divisible by e_{d-n}. In the middle block, first-row column a is proportional to the original diagonal entry at $(d-a, 0)$; it vanishes for $a > d-n$ because $n > d-a$. A polynomial of degree $r < N$ in a cyclic tridiagonal block has a nonzero first-row entry in column r from its unique outermost path. Thus its degree is at most $d-n$. The extreme path (5.13), the duality ratio, and (5.11) determine the sole coefficient:

$$\mathscr{T}_{x,y}(e_n) = \frac{P_n(-s)}{P_{d-n}(-s)} e_{d-n} = \frac{W_n(s)}{H_N(s)} e_{d-n}. \tag{5.14}$$

Swapping x, y gives the other opposite-color endomorphism turn.

For the equal-color orbit, two applications of the entry formula have coefficient

$$\frac{\gamma_a(x)\gamma_k(x)}{\gamma_b(x)\gamma_j(x)} A^{d-k,d-a}_{d-b,d-j}.$$

This is the dual endomorphism with reversed tensor order. The antipode fixes Ω by the two expressions (3.1). Hence a double turn is the same fixed polynomial on the dual equal-color

pair. The second turn is the inverse of the first. There remains one carrier scalar when the opposite reference order is changed, which we now compute rather than normalize away.

Temporarily take generic highest weights $\alpha = d + s + \epsilon/2$, $\beta = d - s + \epsilon/2$. Normalize the degree-$j$ highest vector by its first coefficient. Its last coefficient is

$$b_j = (-1)^j q^{-\beta j + j(j-1)} \prod_{r=1}^{j} \frac{[\beta - r + 1]}{[\alpha - r + 1]}.$$

Only the zero-transfer term on $v_j \otimes w_0$ contributes to the first coefficient after the checked crossing. Its scalar is $q^{(\alpha-2j)\beta/2} b_j$. Multiplying the two orders cancels the factorial ratios and gives

$$q^{\alpha\beta - 2j(\alpha+\beta) + 2j(j-1)}.$$

Replace both weights by $d + \epsilon/2$. Every Casimir node stays fixed; the original exponent is the equal-weight exponent minus s^2. The generic identity therefore extends to $\epsilon = 0$ as a matrix identity, including normals: the canonical coefficients are recovered by the nondegenerate moment pairing (4.2). Thus

$$\check{R}_{y,x}\check{R}_{x,y} = q^{-s^2} r_0(\Omega)^2, \qquad G'G_B = q^{-s^2} I, \quad G' = \check{R}_{y,x} r_0^{-1}. \tag{5.15}$$

Converting the sign-reversed natural reference into the fixed reference (5.2) consequently multiplies coefficients by q^{s^2}. This determines the other two turns.

Domain	Codomain	Scalar γ in (1.4)
$\mathrm{End}(V_+ \otimes V_+)$	$\mathrm{Hom}(B, A)$	$q^{-s^2/2}$
$\mathrm{Hom}(B, A)$	$\mathrm{End}(V_- \otimes V_-)$	$q^{s^2/2}$
$\mathrm{End}(V_- \otimes V_-)$	$\mathrm{Hom}(A, B)$	$q^{s^2/2}$
$\mathrm{Hom}(A, B)$	$\mathrm{End}(V_+ \otimes V_+)$	$q^{-s^2/2}$
$\mathrm{End}(A)$	$\mathrm{End}(B)$	1
$\mathrm{End}(B)$	$\mathrm{End}(A)$	1

Table 1: The six physical forward turns. Here $A = V_+ \otimes V_-$ and $B = V_- \otimes V_+$; every morphism uses the carriers (5.2).

Since $W_k W_{d-k} = H_N^2$, the common map $H_N^{-1}\mathcal{R}\,\mathrm{diag}(W_k)$ is an involution. Together with the listed reciprocal scalars this proves the asserted inverse turns as well as every forward turn.

6 Crossings, their normals and ordinary circles

Proposition 6.1 (The actual crossing polynomial)**.** *On $V_+ \otimes V_+$ the checked crossing has a unique canonical polynomial $R_s \in \mathcal{A}_s$, holomorphic for $|s| < 1$. At generic nodes its values are*

$$R_s(c_j(s)) = \kappa_s (-1)^j q^{j(j+1) - 2sj}, \qquad \kappa_s = q^{(d+s)^2/2}. \tag{6.1}$$

Its inverse and every integer power are holomorphic in the same algebra. On $V_- \otimes V_-$ replace s by $-s$.

Proof. The crossing is an intertwiner because the input colors agree. Lemma 3.2 gives the polynomial and (4.2) proves its coefficient holomorphy, including $s = 0, \pm 1/2$. The highest-vector computation before (5.15), with $\alpha = \beta = d + s$, gives $(-1)^j q^{(d+s)^2/2 - 2(d+s)j + j(j-1)}$. Since $q^{-2Nj} = 1$, this equals (6.1). These values never vanish. A polynomial in a split finite algebra is a unit precisely when it is nonzero at each distinct root: at a double root the inverse has value R^{-1} and derivative $-R'/R^2$. Alternatively apply (4.2) to the holomorphic physical matrix inverse. This proves the assertion for all negative powers. □

For completeness the normal is explicit. Suppose $s_* = 0, 1/2$ or $-1/2$, $j < l$, and $j+l+1-2s_* = N$. The two crossing values agree, since their ratio is $(-1)^{l-j}q^{N(l-j)} = 1$. Their s derivatives differ by $-2\mathrm{i}\theta(l-j)R_s(c_j(s))$; the node derivatives differ by $-8\theta\sin((2j+1-2s_*)\theta)$. Therefore

$$(R^n_{s_*})'(c_j(s_*)) = \frac{\mathrm{i}n(l-j)}{4\sin((2j+1-2s_*)\theta)}R_{s_*}(c_j(s_*))^n, \qquad n \in \mathbb{Z}. \tag{6.2}$$

The prime is the polynomial derivative. The coefficient derivatives cancel in the divided difference of the single holomorphic polynomial $R^n_s(z)$, which justifies this limiting calculation. A reflection-fixed endpoint node is simple, so it has no prescribed normal. In particular at odd N and $s = 0$, the simple node -2 has the value in (6.1) with $j = d/2$. These formulas include both parities and composite levels.

Define the unwrapped angles and the scalar

$$\tau_j(s) = \frac{2\pi}{N}(d/2 + s - j), \quad z_j(s) = -2\cos\tau_j(s) = c_j(s), \quad C_{N,n}(s) = q^{n(d^2/4-s^2/2)}. \tag{6.3}$$

Completing the square gives exactly

$$R^n_s(z_j(s)) = C_{N,n}(s)e^{\mathrm{i}nN\tau_j(s)^2/(4\pi)}.$$

Lagrange interpolation at distinct roots yields, for any polynomial F, with the full product reduced in the actual quotient,

$$\phi_s(R^n_s F) = \frac{(-1)^d N C_{N,n}(s)}{\sin(2\pi s)}\sum_{j=0}^{d}\sin\tau_j(s)F(z_j(s))e^{\mathrm{i}nN\tau_j(s)^2/(4\pi)}. \tag{6.4}$$

Indeed $P_s'(z_j) = N\sin(2\pi s)/\sin\tau_j$. The apparent poles on the right are removable because the left side is holomorphic. Their limits give precisely the full Hermite functional (4.3) with (6.2). Taking the nodal values term by term would generally lose this information. Formula (6.4) is finite, not an asymptotic quadrature formula.

An auxiliary circle of integer color $1 \le j \le N$ carries the ordinary j-dimensional highest-weight module. Starting with the ordered factors V_x, V_y, V_j, its actual adjacent-crossing word is $[2, 1, 1, 2]$, applied in that order and with the intermediate factor dimensions retained. Closing the third factor with K_j^{1-N} gives the block braiding around both main strands. The resulting polynomial is

$$L_j(z) = (-1)^{j-1}U_{j-1}(z/2), \qquad U_0 = 1, \quad U_1(z/2) = z, \quad U_{r+1} = zU_r - U_{r-1}. \tag{6.5}$$

Here is a proof for moving main colors. Keep q fixed and first choose generic roots x, y. Their pair splits into the simple nilpotent channels of Proposition 3.1. On a channel of highest weight ν, temporarily close the auxiliary strand with K_j. In one monodromy direction raising terms kill the highest vector; in the return direction a positive lowering term cannot return to its highest coefficient. The two Cartan factors and pivot give

$$\sum_{a=0}^{j-1} q^{(\nu+1)(j-1-2a)} = U_{j-1}((q^{\nu+1} + q^{-\nu-1})/2).$$

The loop intertwines, so this determines its value on every simple channel. All finite crossing denominators have index below N; continuation in x, y therefore proves the full matrix identity at our repeated channels. Finally $K_j^{1-N} = (-1)^{j-1}K_j$ on the ordinary module. This proves (6.5), with no $1/j$ or $1/N$ normalization and without deforming a moving finite module to generic q.

Since $L_N = (-1)^d P_s'/N$, interpolation and continuation give

$$\begin{aligned}\phi_s(L_N F) &= N\sum_{j=0}^{d} F(z_j(s)),\\ \phi_s(L_N R_s^n F) &= NC_{N,n}(s)\sum_{j=0}^{d} F(z_j(s))e^{inN\tau_j(s)^2/(4\pi)}.\end{aligned} \tag{6.6}$$

The repeated roots are counted because of P_s'. This particular identity does not replace general Hermite traces by values.

For a leaf the rth turned coefficient is exactly

$$h_s^n(r,j) = [e_r]\mathscr{T}_s(R_s^n L_j) = \gamma(s)\frac{H_N(s)}{W_r(s)}\phi_s(e_r R_s^n L_j). \tag{6.7}$$

The type determines γ and the input quotient. For an opposite-color input use its polynomial (8.1). Since $\deg(e_r L_j) \le 2N-2$, the same constant-shift division argument as Lemma 4.1 proves the exact pure identity

$$h_s^0(r,j) = \gamma(s)\frac{H_N(s)}{W_r(s)}h_0^0(r,j). \tag{6.8}$$

It includes zero coefficients, rather than asserting a ratio at a zero. For complete ordinary color and $2r < N$, the complete value is $N^2 S_{2r}/S_r^4$, as proved in [1, Section 4]. All other indices and all ordinary colors are retained by the same polynomial formula. No positive-range truncation is made here.

7 Exact contraction of an ordered tree

A strict tree expression puts an actual two-strand coupon p_v at each vertex and a physical turn on each edge inserting a child in its parent. The ordered children and strand directions are part of the data. Coupons can be crossing powers, ordinary-circle polynomials with retained crossings, or the identity. Remove the carrier of any opposite-order morphism by (5.2); compose the polynomials in the correct source algebra. More generally a binary expression permits arbitrary finite ordered stacking and turns. Induction on this expression is the exact compiler: stack compatible carriers and multiply, or apply the appropriate row of Table 1.

For the tree formula let D_v be the number of incident edges at v, distinct from the level degree d. Let $\sigma_v \in \{0, s, -s\}$ be its actual pair type and put

$$\mathcal{V}_v(k_1,\dots,k_{D_v}) = \phi_{\sigma_v}\left(p_v\prod_{a=1}^{D_v} e_{k_a}\right), \qquad 0 \le k_a \le d. \tag{7.1}$$

Reduction is applied after the entire product. Each edge has the scalar γ_e of its original physical turn; these and the carriers are fixed by the typed diagram.

Theorem 7.1 (Tree contraction and boundary degree)**.** *For a typed tree whose last boundary is an endomorphism, the normalized closing value is*

$$\mathcal{Z}_N(s) = \sum_{\boldsymbol{k}\in\{0,\dots,d\}^E}\prod_v \mathcal{V}_v((k_e)_{e\ni v})\prod_{e\in E}\gamma_e(s)\frac{H_N(s)}{W_{k_e}(s)}. \tag{7.2}$$

After retaining any final closing turn, the zero-framed scalar is

$$F_N(q^s) = \vartheta_{d+s}^{-w(D)} H_N(s)\mathcal{Z}_N(s). \tag{7.3}$$

All physical crossing scalars remain in the coupons; every sum uses all N indices. Any root may be chosen for this summation.

For $z = -2\cos t$, its child-to-parent recursion is

$$M_{v\to p}(u) = \gamma_{vp}\,\phi^{z}_{\sigma_v}\left[p_v(z)K_N(t,u;s)\prod_{w\ \text{child of}\ v} M_{w\to v}(t)\right]. \tag{7.4}$$

At the root omit the kernel and apply ϕ_{σ_0} to its coupon times its incoming messages. As a polynomial in $-2\cos u$, every message has degree at most d.

Proof. If $f = \sum f_j e_j$ and $g = \sum g_j e_j$, the fixed pairing gives

$$\phi(f\mathscr{T}g) = \frac{\gamma}{H_N}\sum_{j=0}^{d} W_j f_j g_j.$$

Put $f(k) = \phi(fe_k) = f_{d-k}$ and similarly $g(k)$ for the covariant legs. Since $W_{d-k}/H_N = H_N/W_k$, this becomes

$$\phi(f\mathscr{T}g) = \sum_{k=0}^{d}\gamma\frac{H_N}{W_k}f(k)g(k). \tag{7.5}$$

This proves both the equal edge indices and their inverse W_k. The respective source and target quotients have the same fixed pairing; physical carriers telescope before this contraction. Removing one terminal vertex therefore produces exactly (7.4). Induction proves (7.2). The final physical partial closure is (3.6), which contributes one H_N, and (2.5) gives the framing. The kernel's second-variable degree is at most d. All other operations act only in its first variable, proving the output degree regardless of the size or unreduced degrees of the subtree.

Finally (7.2) is a finite sum of fixed tensors, so its indices can be summed in any leaf order. Changing the contraction root keeps the original γ_e, carriers and coupons. It does not reverse all physical turns while assigning new forward scalars. Independence of the actual cut was proved separately following (3.6). □

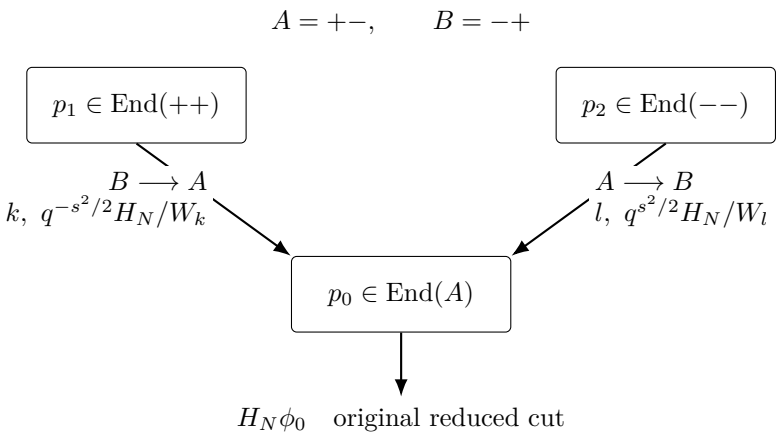


Figure 1: A three-vertex contraction with original types retained. The turns of equal-color coupons p_1, p_2 are maps $B \to A$ and $A \to B$; their ordered product and $p_0 \in \mathrm{End}(A)$ close in $\mathcal{A}_0$. Indices k, l range over $0, \dots, d$. Edge scalars are $q^{-s^2/2}$ and $q^{s^2/2}$, respectively; the additional factors are H_N/W_k and H_N/W_l.

For this example the full unframed value is

$$H_N\sum_{k,l=0}^{d}\frac{H_N}{W_k}\frac{H_N}{W_l}\phi_0(p_0e_ke_l)\phi_s(p_1e_k)\phi_{-s}(p_2e_l).$$

Only the two turn phases cancel; all edge and vertex factors remain. Identity coupons are allowed, so no valence or nonzero-exponent condition has entered the result.

8 The scalar ledger, including negative mixed words

The preceding contraction already includes all physical scalars. We isolate a useful scalar when equal-color chirps are separated from their normalized values and mixed words use r_0^n. This is an exact bookkeeping identity, not a relative estimate for the whole invariant.

Equation (5.15) gives the mixed generators:

	positive crossing	negative crossing
start A	r_0	$q^{s^2} r_0^{-1}$
start B	$q^{-s^2} r_0$	r_0^{-1}.

Each generator swaps order, also for a negative crossing. For a word of length L and exponent n, the exponent of q^{s^2} starting at A is $(L-n)/2 - \lfloor L/2 \rfloor = -\lfloor n/2 \rfloor$; starting at B uses the ceiling. Hence

$$p_{n,A} = q^{-\lfloor n/2 \rfloor s^2} r_0^n, \qquad p_{n,B} = q^{-\lceil n/2 \rceil s^2} r_0^n, \quad n \in \mathbb{Z}. \tag{8.1}$$

These are full quotient identities, including every normal.

Give every endomorphism space the grading $\chi = 0$, and set $\chi(\mathrm{Hom}(A,B)) = 1$, $\chi(\mathrm{Hom}(B,A)) = -1$. It adds under compatible products. Table 1 says each turn scalar is $q^{(\chi_{\mathrm{out}} - \chi_{\mathrm{in}})s^2/2}$. Temporarily put $\widetilde{f} = q^{-\chi s^2/2} f$ in each type. Products still multiply, turns lose precisely their scalar, and the final endomorphism is unchanged. A mixed word has $\chi = \delta(n)$ starting at A and $-\delta(n)$ starting at B, where $\delta(n)$ is Euclidean parity, including when $n < 0$. Both formulas (8.1) then leave $q^{-ns^2/2}$. For equal-color words the same scalar is $C_{N,n}(s)/C_{N,n}(0)$ in (6.3); their remaining chirps are still evaluated at shifted angles.

Proposition 8.1 (Exact quadratic scalar)**.** *Let the actual checked-crossing powers in the original diagram be n_i, and let $w(D)$ be its oriented writhe. Exclude the index-dependent factors H_N/W_k, the single closing factor H_N, and the shifted chirp or fixed $r_0^{n_i}$ at each box. The product of the remaining scalar ratios is*

$$A_N(s) = q^{(-w(D) - \sum_i n_i)s^2/2}. \tag{8.2}$$

For the positive checked-braid convention, put $o_i = 1$ when the original oriented strands in box i are parallel, $o_i = -1$ when antiparallel. Then $w(D) = \sum_i o_i n_i$ and the exponent integer is $-2\sum_{o_i=1} n_i$.

Proof. The grading leaves $q^{-n_i s^2/2}$ from every box. Equation (2.5) adds $q^{-w(D)s^2/2}$. Bending maps have already been counted in the turns, and their snake identities leave no further monomial; the closure leaves the one excluded H_N. Multiplying proves the formula. Reversing one oriented crossing tangent reverses the writhe sign, whereas reversing both preserves it, proving the final assertion. □

For example an exponent -1 word starting at A has scalar q^{s^2} before the turn grading, whereas one starting at B has scalar one. Neither conversion is counted twice. The ledger does not say that a whole polynomial or invariant is its complete value times $A_N(s)$: moving arguments and changed quotient products remain inside the exact expression.

9 Finite reproduction and interfaces

The accompanying `check.py` compares physical module matrices, finite crossing sums and explicit currying contractions with these polynomial formulas in exact finite fields. Its report lists all levels, rational meridian parameters and primes. It checks both parities, collisions, negative powers, all six turns, full pivotal moments, curls, moving ordinary circles and a small tree. These finite checks are supplementary to the all-level coefficient and path proofs. Run

```
python3 papers/meridian-tangles/check.py -output DIRECTORY.
```

Python and NumPy are required. The script rejects disabled assertions and removes an old success record before checking.

The finite input to an analytic tree theorem is now explicit: (7.1) is its actual local tensor, (6.4) its full trace, (1.5) its edge kernel and (7.4) its original recursion. An asymptotic theorem must estimate the whole contraction, including complementary Newton indices and repeated-root normals. None of the identities above establishes dominance of a selected saddle or positive subcollection.

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